Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/27/2005
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Project Title:	The Effects of Vector-Averaged Gravity on the Development of T cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:	PI is deceased as of 10/26/2007 (info from A. Chu/ARC; 9/16/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC97G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have developed a microgravity organ culture dish system (MOCDS) that allows for the normal development of T cells from hematopoietic stem cell (HSC) progenitors. This system mimics the normal development of T cells outside of the body, away from stress hormones made during flight. The MOCDS can be used both on the clinostat and during spaceflight. The culture systems are easy to set up and manipulate which allows for a more detailed analysis. Using this specialized system, our work has revolved around studying the effects of vector-averaged gravity on the development of T cells. In our previous work that utilized MOCDS we demonstrated the gravity-dependence of T cell development. The results from this work suggest that the full sequelae of pre-TCR signaling may not be taking place in vector-averaged gravity, as indicated by a block at the immature single positive stage of development. Since then, we have set out to determine what the underlying mechanism was for this block. Our current focus has been on analyzing the cytokine profiles for our clinorotated cultures, specifically TNF-alpha. Using the information obtained from these		

	and our other studies, we aim to better understand, and hopefully reverse, the effects of clinorotation. Determine the mechanism by which vector-averaged gravity blocks T cell development at the ISP developmental stage using organ culture, which preserves the 3-D microenvironment of the tissue. This will allow for better understanding of the mechanism underlying decreased T cell development in vector-averaged gravity. This knowledge can be applied to reversing the effects of vector-averaged gravity on developing T cells. A functional immune system is critical if humans intend to reside in environments that have minimal gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work has shed light on the signal dampening effects of microgravity exposure. By further understanding the underlying mechanism of this microgravity-induced effect, we could initiate a new line of research centered on the treatment and prevention of other thymocyte developmental problems, such as Type I diabetes. Many T cell developmental problems involve signaling defects similar to those seen in our microgravity work. For instance, microgravity exposure led to a significant increase in the production of TNF-alpha. Other work investigating the role of TNF-alpha during T cell development has shown that over expression of this cytokine may lead to deficient negative selection and hence the production of potentially autoreactive T cells.
Task Progress:	The article describing the majority of our work is currently in press and will soon be published in Developmental and Comparative Immunology. Generally, this work shows that after 4 days of vector-averaged gravity exposure, the fetal thymus organ cultures (FTOC) exhibited a significant reduction in CD4+CD8+ double positive (DP) cell production. Furthermore, CD4-CD8- double negative (DN) cells expressed lower levels of CD3 on their surface. DN, as well as immature single positive (ISP) cells, also expressed reduced levels of the IL-7 receptor alpha chain (CD127). These reductions in CD3 and CD127 expression were concomitantly associated with an increased production of tumor necrosis factor (TNF)-alpha. We were also able to show that addition of an exogenous signal (anti-CD3 epsilon monoclonal antibody) to these cultures effectively mitigated the MMG-induced effects, therefore suggesting that MMG-exposure, does in fact, cause a signal dampening effect on developing thymocytes. Currently our focus has been extended to further analyze the increased production of tumor necrosis factor (TNF)-alpha in MMG. Its increased presence in clinorotated cultures suggests that it plays an important role in the development of DP thymocytes. To analyze further, we added soluble TNF receptor one (TNFR1) (1ug) to three day cultures and found that double positive cell production was increased in both the clinostat and motional control. More specifically, the addition of the soluble receptor to the clinorotated culture was able to increase the level of DP thymocytes and decrease the level of DN thymocytes to a level comparable to the motional control without TNF receptor. Interestingly, the motional control showed a higher DN population compared to the DP thymocyte population, but with the addition of TNF receptor, these values changed to levels similar to the stationary control. This suggests that the vibration of the motional control does have an intermediate effect, lying somewhere between the clinostat and the stationary control, but the clinostat clearly has an additional block presumably due to the vector-averaged gravity exposure. We also attempted to mimic the clinostat effect by adding 15 pg TNF-a to the stationary control culture. This level of TNF is comparable to that typically seen in clinorotated cultures. The results indicated that despite the addition of TNF-a, the frequency DN and DP thymocyte levels did not change from that seen in the other stationary controls. However, there was a marked decrease in overall cell yield. Clearly, there must be other factors affecting T cell development in these cultures. To further analyze the effects of TNF-alpha, we used confocal microscopy to visualize TNF-alpha production in our cultures. The motional and stationary controls did not show any TNF-alpha staining, on the other hand, the clinorotated cultures did exhibit profuse TNF-alpha staining throughout the lobes. The isotype control used in this experiment also showed staining in the clinorotated lobes, but again not in the motional and stationary lobes. However, this data was derived from a single experiment and will have to be reproduced in order to validate this initial observation. On a side note, we noticed that cells in the motional and stationary control showed more cell spreading onto the filter substrate compared to the clinostat which showed minimal spreading. These results suggest not only the decreased cellularity of clinorotated cultures which has been seen in our previous experiments, but also the possibiligy of an inability of the developing thymocytes to properly migrate.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Articles in Peer-reviewed Journals	Woods CC, Banks KE, Gruener R, DeLuca D. "Loss of T cell precursors after spaceflight and exposure to vector-averaged gravity." FASEB J. 2003 Aug;17(11):1526-8. Epub 2003 Jun 3. PubMed PMID: 12824295 , Aug-2003
Articles in Peer-reviewed Journals	Woods CC, Banks KE, Lebsack TW, White TC, Anderson GA, Maccallum T, Gruener R, DeLuca D. "Use of a microgravity organ culture dish system to demonstrate the signal dampening effects of modeled microgravity during T cell development." Developmental and Comparative Immunology, Currently in Press, as of March 2005. , Mar-2005